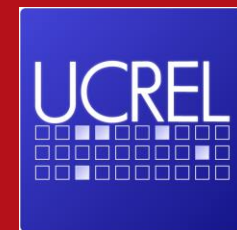


UCREL Summer School Hackathon – Multilingual Named Entity Recognition

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Multilingual Named Entity Recognition (NER)

Dr. Bob referred the patient to the Lakeside Hospital in Manchester.

Dr. Bob – Person

Lakeside Hospital – Organisation

Manchester – Location

Applications of NER

- **Information Extraction:** Identifies key entities (e.g., people, organizations, locations, dates) from unstructured text, converting it into structured information.
- **Search & Information Retrieval:** Improves search engines by understanding entities in queries and documents, leading to more accurate search results.
- **Question Answering & Chatbots:** Helps AI assistants recognize important entities in user queries to provide precise and context-aware responses.
- **Knowledge Graph Construction:** Extracts entities and their relationships to build structured knowledge bases used by search engines, recommendation systems, and AI applications.

Multilingual NER - Data

A subset of the Universal NER (v1 and v2) datasets; [Mayhew et al. \(2024\)](#) and [Blevins et al. \(2026\)](#)

- Only contains Person (PER), Organisation (ORG), and Location (LOC) labels.

Language	Domain
English	Web media; blogs, news, email, reviews and Yahoo! answers
Danish	News wire and social media
Simplified Chinese	Wikipedia
Surprise	?

Suggested approaches

- supervised model training;
- fine-tuning an existing multilingual model;
- Training with or without synthetic-data augmentation;
- few-shot or zero-shot prompting with a large language model;
- a combination of these approaches.

Suggested approaches

- Existing NER models that might be of interest:
 - <https://huggingface.co/Babelscape/wikineural-multilingual-ner>
 - [knowledgator/gliner-x-small · Hugging Face](#)
- Small LLMs... on [HuggingFace](#) can be ran on Google Colab
- Large LLMs... [OpenRouter](#) sometimes have free model credits for certain LLMs

Prizes

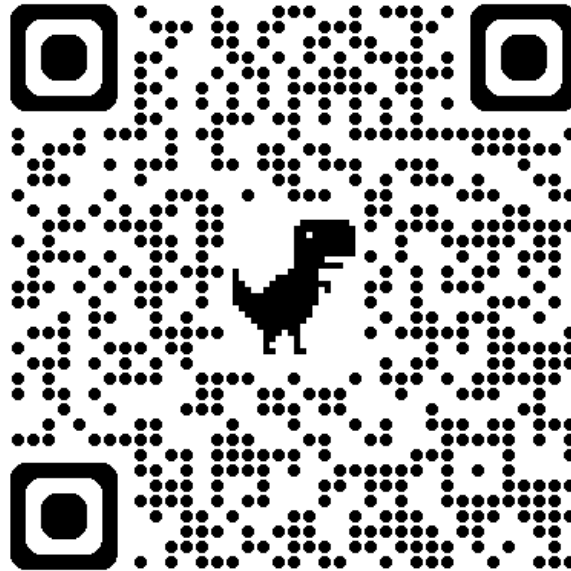
- **Best overall performance** across the three disclosed languages.
- **Best performance on the surprise language.**
- **Most innovative solution**, recognising originality rather than predictive performance alone.

Creating Teams.....

- We are suggesting teams of around 4 people.
- Choose your teams wisely.

Practical activities

GitHub - <https://github.com/UCREL/USS-Hackathon-Starter-Code>



Practical activities

CodaBench-

[https://www.codabench.org/competitions/17588/
?secret_key=de718acd-a2df-4dc6-b486-
65a9f227d698](https://www.codabench.org/competitions/17588/?secret_key=de718acd-a2df-4dc6-b486-65a9f227d698)



References

- [Universal NER: A Gold-Standard Multilingual Named Entity Recognition Benchmark](#) (Mayhew et al., NAACL 2024)
- [Universal NER v2: Towards a Massively Multilingual Named Entity Recognition Benchmark](#) (Blevins et al., LREC 2026)

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